

SIMATS ENGINEERING

Name	Register Number	Course Code	Course
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15. Perform transformation using Direct Linear Transformation.

AIM:

Transformation using Direct Linear Transformation.

PROGRAM:

```
import cv2
import numpy as np

img = cv2.imread(r'C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png')

if img is None:
    print("Error: Image not found!")
else:
    rows, cols = img.shape[:2]

    # Source points (corners of the original image)
    src_points = np.float32([
        [0, 0],
        [cols - 1, 0],
        [0, rows - 1],
        [cols - 1, rows - 1]
    ])

    # Destination points (apply DLT transformation)
    dst_points = np.float32([
        [50, 50],
        [cols - 50, 30],
        [30, rows - 50],
        [cols - 30, rows - 30]
    ])

    # Compute transformation matrix using Direct Linear Transformation (DLT)
    H, status = cv2.findHomography(src_points, dst_points)

    # Apply the transformation
    dlt_img = cv2.warpPerspective(img, H, (cols, rows))
```

```
# Save the output
cv2.imwrite(
    r'C:\Users\Susmitha\Downloads\ITA0510 Lab\DLT_Transformed_Image.jpg',
    dlt_img
)

# Display the result
cv2.imshow("Original Image", img)
cv2.imshow("DLT Transformed Image", dlt_img)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

INPUT:



OUTPUT:



RESULT:

Thus the program is executed successfully.